

## 0.1 Topological Maps

**Definition 0.1.1.** Let  $X, Y$  be Topological Spaces. A map  $f: X \rightarrow Y$  is said to be *continuous* at  $x_0 \in X$  if:

For any open  $V \in \mathcal{T}_Y$  with  $f(x_0) \in V$ , there is an open  $U \in \mathcal{T}_X$  with  $x_0 \in U$  such that  $f(U) \subset V$

If  $f$  is continuous at every point in the domain, then  $f$  is called a *continuous map*.

**Definition 0.1.2.** Let  $X, Y$  be Topological Spaces.

A map  $f: X \rightarrow Y$  is called an *open map* if: For any open  $U \subseteq X$ ,  $f[U] \subseteq Y$  is open in  $Y$ .

A map  $f: X \rightarrow Y$  is called a *closed map* if: For any closed  $C \subseteq X$ ,  $f[C] \subseteq Y$  is closed in  $Y$ .

**Theorem 0.1.3.** Let  $f: X \rightarrow Y$  be a map. Then, The Followings are Equivalent:

- a)  $f$  is Continuous Map.
- b) For any closed  $C \subset Y$ ,  $f^{-1}[C] \subset X$  be closed.
- c) For any subset  $A \subset X$ ,  $f[\overline{A}] \subset \overline{f[A]}$ .
- d) For any subset  $B \subset Y$ ,  $f^{-1}[A^\circ] \subset (f^{-1}[B])^\circ$ .

*Proof.*

a)  $\implies$  b) Let  $C \subset Y$  is closed. Then,

$$f^{-1}[Y \setminus C] = X \setminus f^{-1}[C]$$

is open in  $Y$ , thus  $f^{-1}[C]$  is closed.

b)  $\implies$  c) Let  $A \subset X$  be given. Then,

$$A \subset f^{-1}[f[A]] \subset \underbrace{f^{-1}[f[A]]}_{\text{closed by b)}} \implies \overline{A} \subset f^{-1}[\overline{f[A]}] \implies f[\overline{A}] \subset f[f^{-1}[\overline{f[A]}]] \subset \overline{f[A]}$$

c)  $\implies$  d) Let  $B \subset Y$ , put  $A = f^{-1}[Y \setminus B] = X \setminus f^{-1}[B]$ . Then,

$$f[\overline{A}] = f[\overline{X \setminus f^{-1}[B]}] = f[X \setminus (f^{-1}[B])^\circ] \text{ and } \overline{f[A]} = \overline{f[f^{-1}[Y \setminus B]]} \subset \overline{Y \setminus B} = Y \setminus B^\circ$$

By c),

$$\begin{aligned} f[\overline{A}] &= f[X \setminus (f^{-1}[B])^\circ] \subset \overline{f[A]} \subset Y \setminus B^\circ \\ \implies X \setminus (f^{-1}[B])^\circ &\subset f^{-1}[f[X \setminus (f^{-1}[B])^\circ]] \subset f^{-1}[Y \setminus B^\circ] = X \setminus f^{-1}[B^\circ] \\ \implies f^{-1}[B^\circ] &\subset (f^{-1}[B])^\circ \end{aligned}$$

d)  $\implies$  a) Let  $U \subset Y$  be an open set. By d),

$$f^{-1}[U] \stackrel{U \text{ open}}{=} f^{-1}[U^\circ] \subset (f^{-1}[U])^\circ$$

Meanwhile, reverse inclusion is clear, thus  $f^{-1}[U] = (f^{-1}[U])^\circ$ , open. □

**Lemma 0.1.4.** Let  $X, Y$  be Topological Space. Then,

1.  $f: X \rightarrow Y$  is open map if and only if For any  $A \subset X$ ,  $f[A^\circ] \subset (f[A])^\circ$ .
2.  $f: X \rightarrow Y$  is closed map if and only if For any  $A \subset X$ ,  $\overline{f[A]} \subset f[\overline{A}]$ .

**Lemma 0.1.5.** Let  $X, Y$  be Topological Space, and  $f : X \rightarrow Y$  be a bijection. Then, TFAE:

1.  $f$  is open map.
2.  $f$  is closed map.
3.  $f^{-1} : Y \rightarrow X$  be continuous map.

Clearly, Homeomorphism is open, closed, continuous map.

**Lemma 0.1.6.** Let  $f : X \rightarrow Y$  be a Homeomorphism,  $A \subset X$ . Then, followings hold:

1.  $f[\overline{A}] = \overline{f[A]}$ .
2.  $f[A^\circ] = (f[A])^\circ$ .

*Proof.* 1. is clear by  $f$  is continuous, and closed map.

2.  $\subset$ )  $A^\circ \subset A \implies f[A^\circ] \subset f[A] \implies f[A^\circ] \subset (f[A])^\circ \subset f[A]$  by  $f$  open map.

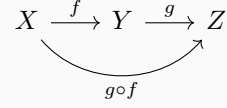
2.  $\supset$ ) Let  $x \in (f[A])^\circ$  be given. Then, there is an open  $U \in \mathcal{T}$  such that  $x \in U \subset f[A]$ .

Now,  $f^{-1}[x] \in f^{-1}[U] \subset f^{-1}[f[A]] = A$  by  $f$  is bijection, this implies

$(f^{-1}[U])^\circ = f^{-1}[U] \subset A^\circ \implies f^{-1}[x] \in f^{-1}[U] \subset A^\circ \implies f[f^{-1}[x]] = x \in f[A^\circ]$ . □

**Theorem 0.1.7.** Let  $X, Y, Z$  be Topological Spaces, and  $f : X \rightarrow Y, g : Y \rightarrow Z$  be functions.

1. If  $f, g$  are Continuous map, then  $g \circ f$  is Continuous map.
2. If  $f, g$  are Open map, then  $g \circ f$  is Open map.
3. If  $f, g$  are Closed map, then  $g \circ f$  is Closed map.



Above three statements are trivial.

1. If  $g \circ f$  is Open map,  $f$  is Continuous onto map. Then,  $g$  is Open map.
2. If  $g \circ f$  is Open map,  $g$  is Continuous one-to-one map. Then,  $f$  is Open map.

*Proof.* 1) Let  $U \in \mathcal{T}_Y$  be an open set. Since  $f$  is Continuous map,  $f^{-1}[U]$  is open of  $X$ .

Now,

$$\underbrace{(g \circ f)[\underbrace{f^{-1}[U]}_{\text{open}}]}_{\text{image of open map}} = g[f[f^{-1}[U]]] \stackrel{\text{onto}}{=} g[U] \in \mathcal{T}_Z$$

2) Let  $U \in \mathcal{T}_X$  be an open set. Since  $g \circ f$  is Open map,  $(g \circ f)[U]$  is open of  $Z$ .

Now, by  $g$  is Continuous one-to-one,

$$g^{-1}[(g \circ f)[U]] = g^{-1}[g[f[U]]] \stackrel{1 \text{ to } 1}{=} f[U] \in \mathcal{T}_Y$$

□

**Lemma 0.1.8. Pasting Lemma**

Suppose that  $X, Y$  are Topological Space, and  $A, B \subset X$  such that  $X = A \cup B$ .

If both  $A, B$  are Open or Closed,  $f : A \rightarrow Y$  and  $g : B \rightarrow Y$  are Continuous map such that  $f|_{A \cap B} = g|_{A \cap B}$ , then

$$f \cup g : X \rightarrow Y : \begin{cases} f(x) & x \in A \\ g(x) & x \in B \end{cases}$$

is Continuous map.